MATH 223: Linear Algebra

William Homier[1]

[1] *McGill University Physics, 3600 Rue University, Montréal, QC H3A 2T8, Canada*

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Abstract

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1 Introduction

2 Prerequisite knowledge

2.1 Introduction to Proof

2.1.1 Direct Proof

The basic idea of a direct proof of $P \implies Q$ is:

1. Assume P is true,
2. Use P to show that Q is true (the hard part).

Example. *An integer number n is odd if and only if there exists an integer k such that $n = 2k + 1$.*

Proof. Assume a, b are consecutive integers, meaning $b = a + 1$. So

$$a + b = a + (a + 1) = 2a + 1$$

Thus b is odd if and only if a is odd. □

2.1.2 Proof by Contradiction

The idea here is that a proposition is either true or false, but not both. We get a contradiction when we can show a statement is both true and false, showing our initial assumptions are inconsistent. We use this to show $P \implies Q$ by assuming both P and not Q are simultaneously true and deriving a contradiction. Hopefully, we have either proved or assumed P to be true so that by default, not Q is false.

Example. *An integer number n is odd if and only if there exists an integer k such that $n = 2k + 1$.*

Proof. Assume a and b are consecutive integers. Assume that $(a + b)$ is not odd. If $(a + b)$ is not odd, then no integer k such that $(a + b) = 2k + 1$. But $a + b = a + (a + 1) = 2a + 1$. Shown that $(a + b) \neq 2k + 1$ but $(a + b) = 2a + 1$, showing a contradiction. □

2.1.3 Proof by Contrapositive

The idea behind a proof by contrapositive is to prove $P \implies Q$ by treating it the other way around meaning proving not $Q \implies$ not P.

Example. *If n^2 is even, then n is even.*

Proof. To prove this statement, we need to follow the proof by contrapositive, which states that we need to show if n is not even, n^2 is not even too. Assume n is odd, $\exists k_1 \in \mathbb{Z}, n = 2k_1+1$. So $n^2 = (2k_1+1)^2 = 4k_1^2+4k_1+1 = 2(2k_1^2+2k_1)+1$. Let $k_2 = 2k_1^2 + 2k_1$. So $n^2 = 2k_2 + 1$. Thus n^2 is odd. □

2.2 Notation

2.2.1 Sets

Sets are a grouping of objects.

Set	Meaning	Examples
$\mathbb{N}$	The set of natural numbers	(0, 1, 2, 3, ...)
$\mathbb{Z}$	The set of integers	(..., -3, -2, -1, 0, 1, 2, 3, ...)
$\mathbb{Q}$	The set of rational numbers	$\mathbb{Q} = \frac{a}{b} \mid \forall a, b \in \mathbb{Z}\ and\ b \neq 0$
$\mathbb{R}$	The set of all rational and all irrational numbers	$(..., -1, 0, \frac{1}{4}, 1, 1000, ...)$
$\mathbb{C}$	The set of all complex numbers	$\mathbb{C} = \{x + iy \mid x, y \in \mathbb{R}\ and\ i \subseteq \sqrt{-1}\}$.

We have the following relationships between sets:

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$$

2.2.2 Symbols

Symbol	Meaning
$\subseteq$	is a subset of or equal to
$\subset$	is a strict subset of
$\in$	is an element of
$\forall$	for all
$\exists$	there exists
$\emptyset$	empty set
$\Rightarrow$	implies
$\Leftrightarrow$	if and only if
$\cong$	is isomorphic to

[1]

2.3 Familiarity with $\mathbb{R}^n$

Definition 1 ($\mathbb{R}^n$)**.** *Let $n \in \mathbb{N}$. The Cartesian product of n copies of $\mathbb{R}$ is called $\mathbb{R}^n$.*

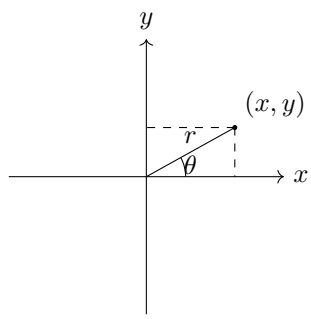
$$\mathbb{R}^n = \{ \begin{pmatrix} x_1 \\ \dots \\ x_n \end{pmatrix} : x_1, \dots, x_n \in \mathbb{R} \}$$

2.4 Polar Coordinates

Definition 2 (Polar coordinates)**.** *Instead of describing a point by (x, y), we may describe it using polar coordinates (r, θ), where r is the distance to the origin and θ is the angle with the positive x-axis.*

[1] is isomorphic to = structurally the same as...

Example. *Consider the point (x, y), where $x = r\cos(\theta)$ and $y = r\sin(\theta)$. We can define (r, θ) as follows:*



2.5 Complex Algebra

2.5.1 Complex Numbers

Definition 3 (Complex Number)**.** *A complex number is of the form: $z = x + iy$ where $x, y \in \mathbb{R}$ and i is the imaginary unit $i = \sqrt{-1}$.*

Theorem 1 (Fundamental Theorem of Algebra)**.** *The FTA states that any non-constant, single-variable polynomial*[2] *with complex coefficients has at least one root in $\mathbb{C}$.*

Remark 1. *If we have a polynomial f of degree n, then it has n roots, where each root can have a multiplicity*[3]*.*

Example. *If we have a polynomial $(x-1)^2$, it has a degree of 2 but only one root, which is 1, with a multiplicity of 2.*

We can factorize a polynomial in the form of $f = a_n x^n + ... + a_1 x + a_0$ into a linear factor: $f = a(x - z_1)(x - z_2)...(x - z_n)$ where z_i are the roots of f in $\mathbb{C}$. Therefore, the FTA implies that f has a root $f(z) = a(z - z) = 0$.

2.5.2 Complex Operations

We can define operations on complex numbers as follows:

- Addition: $z + z' = (x + x') + i(y + y')$, where $x, x', y, y' \in \mathbb{R}$.
- Multiplication: $zz' = (x + iy)(x' + iy') = (xx' - yy') + i(xy' + yx')$.
- Inverse: $\frac{1}{z} = \frac{\overline{z}}{z\overline{z}} = \frac{x-iy}{x^2+y^2} = \frac{x}{x^2+y^2} + i\frac{-y}{x^2+y^2}$

Multiplying by a complex number z corresponds geometrically to

$$\begin{cases} \text{a rotation by some angle } \theta, \\ \text{a rescaling by the factor } |z|. \end{cases}$$

[2] Polynomial is a function such as: $f = a_n x^n + a_{n-1}x^{n-1} + ... + a_1 x + a_0$ where $a_i \in \mathbb{R}$ or $\mathbb{C}$ and $n \in \mathbb{N}$.

[3] The multiplicity of a root represents how many times the root occurs in the polynomial.

2.5.3 Complex Conjugate

Definition 4 (Complex Conjugate)**.** *A complex conjugate is a way to "flip" the imaginary part of a complex number. For example, if we have a complex number $z = x + iy$, then the complex conjugate of z is $\overline{z} = x - iy$.*

Some basic properties of complex conjugates are:

- $\overline{\overline{z}} = z$
- $\overline{z + z'} = \overline{z} + \overline{z'}$
- $\overline{z \cdot z'} = \overline{z} \cdot \overline{z'}$

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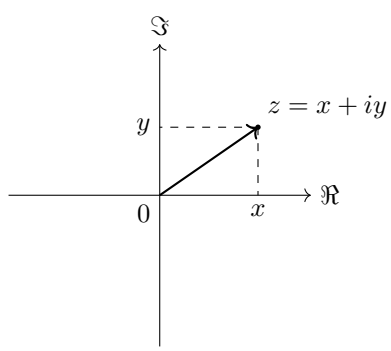
2.5.4 Geometric Interpretation of Complex Numbers

Definition 5 (Geometric interpretation)**.** *Every complex number $z = x + iy$ can be identified with a point (x, y) in the plane, called the complex plane.*

We define the complex plane as:

$$\mathbb{C} = \{x + iy \mid x, y \in \mathbb{R}\}.$$



We can rewrite the definition of the unit circle as follows:

$$S' = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\} = \{z \in \mathbb{C} : |z| = 1\},$$

where S' is the unit circle in the complex plane.

2.5.5 Modulus

Definition 6 (Modulus)**.** *The modulus of a complex number z is defined by*

$$|z| = \sqrt{z\overline{z}} = \sqrt{x^2 + y^2}.$$

Geometrically, $|z|$ is the distance from the origin to the point (x, y). Recall r in polar coordinates, which is the same as $|z|$.

2.5.6 Polar Form of Complex Numbers

Definition 7 (Euler's formula)**.**

$$e^{i\theta} = \cos\theta + i\sin\theta \tag{1}$$

Definition 8 (Polar and exponential form)**.** *If $z = x + iy$ with $r = \sqrt{x^2 + y^2}$, then*

$$x = r\cos\theta, \quad y = r\sin\theta,$$

so

$$z = r(\cos\theta + i\sin\theta) = re^{i\theta}.$$

Definition 9 (Multiplication in polar form)**.**

$$z = re^{i\theta}, \quad z' = r'e^{i\theta'}, \quad zz' = rr'e^{i(\theta+\theta')}.$$

Example.

$$(1+i)^{32} = (\sqrt{2}e^{i\pi/4})^{32} = (\sqrt{2})^{32}e^{i8\pi} = 2^{16}(\cos 8\pi + i\sin 8\pi) = 2^{16}.$$

2.5.7 Roots in $\mathbb{C}$

Definition 10 (n^{th} roots in $\mathbb{C}$)**.** *For any complex number z, an nth root of z is a complex number w such that*

$$w^n = z.$$

Example. *If $z = re^{i\theta}$, then any solution of $w^n = z$ must satisfy*

$$w_k = r^{1/n}e^{i(\theta+2\pi k)/n}, \quad k = 0, 1, 2, \ldots, n-1,$$

where the nth roots of z are equally spaced on a circle of radius $r^{1/n}$ centered at the origin.

Definition 11 (Roots of unity)**.** *The nth roots of unity are the solutions of a special case where $z = 1$.*

Example. *If $z = 1$, then any solution of $w^n = z$ must satisfy*

$$w_k = e^{i2\pi k/n}, \quad k = 0, 1, 2, \ldots, n-1.$$

Geometrically, they lie on the unit circle and are equally spaced.

2.5.8 Hermitian

Definition 12 (Hermitian Matrix)**.** *A complex square matrix $A \in M_n(\mathbb{C})$ is called Hermitian if it is equal to its conjugate transpose:*

$$A = A^*$$

where $A^* = \overline{A}^T$.

Equivalently, a matrix $A = (a_{ij})$ is Hermitian if

$$a_{ij} = \overline{a_{ji}} \quad \text{for all } i, j.$$

Example.

$$A = \begin{pmatrix} 2 & 1+i \\ 1-i & 3 \end{pmatrix}$$

is Hermitian since

$$A^* = \begin{pmatrix} 2 & 1+i \\ 1-i & 3 \end{pmatrix} = A.$$

Remark 2. *All eigenvalues of a Hermitian matrix are real and eigenvectors corresponding to distinct eigenvalues are orthogonal.*

Problem 1 (Hermitian Matrix)**.** *Let $A, B \in M_n(\mathbb{C})$. Prove that $(AB)^* = B^*A^*$.*

2.6 Basic Algebraic structures

2.6.1 Sets with Multiplication

Definition 13 (Set with multiplication)**.** *A set M is called a set with multiplication if you can multiply any two elements of M, and the result is still in M. In other words, for any $a, b \in M$, the product ab is also in M.*

Example. *An example of a set with multiplication is the set of all 2×2 complex matrices: $M = M_2(\mathbb{C})$. Another example is the nonzero set of all real numbers $\mathbb{R}$ with ordinary multiplication: $\mathbb{R}^* = \mathbb{R} \setminus \{0\}$.*

Example. *Let $M = \mathbb{R}$. If $a, b \in \mathbb{R}$, then $ab \in \mathbb{R}$. So the real numbers $\mathbb{R}$ form a set with multiplication.*

2.6.2 Invertibility

Definition 14 (Condition for Invertibility)**.** *Let $A \in M$ be an $n \times n$ matrix, and suppose that there exists an $n \times n$ matrix B such that $AB = I_n$ or $BA = I_n$. Where I_n is the $n \times n$ identity matrix*[4] $\begin{bmatrix} 1 & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & 1 \end{bmatrix}$. *Then A is invertible, and B is called the inverse of A and is denoted by $B = A^{-1}$.*

Remark 3. *If A is invertible, then A^{-1} exists and is unique*[5]*.*

To determine if an element A in a set with multiplication M is invertible, we can use the following examples:

[4] An identity matrix is a square matrix with 1s on its main diagonal and 0s everywhere else. It represents no change in linear transformations, and it's used in finding matrix inverses.

[5] Unique means there is exactly one such element.

Example. *Let $M = \mathbb{Z} = \{... - 2, -1, 0, 1, 2, ...\}$ and $A = 2$. Is A invertible in M?*
Solution: No, because $\frac{1}{2} \notin \mathbb{Z}$.

Example. *Let $M = \mathbb{R}$ and $A = 2$, is A invertible in M?*
Solution: Yes, because $\frac{1}{2} \in \mathbb{R}$.

Example. *Is $1 + i$ invertible in $\mathbb{C}$?*
Solution: Yes, using our previous definition of inverse (2.5.2), we get that

$$\frac{1}{1+i} = \frac{1-i}{2} \in \mathbb{C}.$$

Problem 2 (Invertibility)**.** *Show that if an inverse of A in $\mathbb{M}$ exists, then it is unique.*

Problem 3 (Invertibility 2)**.** *Let K be a field. Prove that this matrix $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ $\in M_2(K)$ is not invertible.*

2.6.3 Ring

Definition 15 (Ring)**.** *A ring is a set R where you can add and multiply elements, and the following are true:*

1. *You can add any two elements and stay in R. There is a zero, every element has a negative, and addition is commutative[6] and associative[7].*
2. *You can multiply any two elements and stay in R. Multiplication is associative, and there is a 1.*
3. *Multiplication distributes over addition:*

$$a(b + c) = ab + ac \quad \text{and} \quad (a + b)c = ac + bc.$$

Example. *The set of integers $\mathbb{Z}$ with the usual addition and multiplication is a ring. First, $\mathbb{Z}$ is closed under addition and multiplication. If $a, b \in \mathbb{Z}$, then $a + b \in \mathbb{Z}$ and $ab \in \mathbb{Z}$. There is a zero element 0 such that*

$$a + 0 = a \quad \text{for all } a \in \mathbb{Z},$$

and every integer has an additive inverse since

$$a + (-a) = 0.$$

Addition is commutative and associative:

$$a + (b + c) = (a + b) + c.$$

[6] Property which focuses on changing order of addition, i.e $a + b = b + a$.
[7] Property which focuses on changing grouping of addition, i.e $a + (b + c) = (a + b) + c$.

Multiplication is associative and has identity 1*:*

$$a(bc) = (ab)c,$$

$$a \cdot 1 = a.$$

Finally, multiplication distributes over addition:

$$(a + b)c = ac + bc.$$

Therefore, $(\mathbb{Z}, +, \cdot)$ *is a ring.*

2.6.4 Field

Definition 16 (Field)**.** *A field is a ring* (15) *in which every nonzero element has a multiplicative inverse.*

Example. *The set of rational numbers* $\mathbb{Q}$ *with the usual addition and multiplication is a field. First,* $(\mathbb{Q}, +, \cdot)$ *is a ring because rational numbers are closed under addition and multiplication, addition is commutative and associative, there is a zero element* 0*, and multiplication is associative with identity* 1*. In addition, every nonzero element has a multiplicative inverse. If* $q \in \mathbb{Q}$ *and* $q \neq 0$*, then there exists* q^{-1} *such that*

$$q \cdot q^{-1} = 1.$$

For example,

$$\frac{3}{5} \cdot \frac{5}{3} = 1.$$

Therefore, $(\mathbb{Q}, +, \cdot)$ *is a field.*

Remark 4. *The integers* $\mathbb{Z}$ *form a ring but not a field because most nonzero elements do not have a multiplicative inverse in* $\mathbb{Z}$ *(for example, there is no integer* x *such that* $2x = 1$*).*

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3 Vector Spaces

Definition 17 (Vector Space)**.** *Let* k *be a field. A vector space over* k *(or* K*-vectorspace) is a set* V *with addition and scalar multiplication which satisfies the following axioms*

1. *Closure under addition:* $v_1 + v_2 \in V$
2. *Commutativity of addition:* $v_1 + v_2 = v_2 + v_1$
3. *Associativity of addition:* $(v_1 + v_2) + v_3 = v_1 + (v_2 + v_3)$
4. *Identity Element of Addition: There exists a zero vector* $0 \in V$ *such that* $v + 0 = v$

5. *Existence of additive inverses:* $\forall v \in V, \exists -v \in V$ *with* $v + (-v) = 0$
6. *Closure under scalar multiplication:* $\lambda v \in V$
7. *Compatibility of scalar multiplication with field multiplication:* $\lambda(\mu v) = (\lambda\mu)v$
8. *Identity element of scalar multiplication:* $1v = v$
9. *Distributivity of scalar multiplication over vector addition:* $\lambda(v_1 + v_2) = \lambda v_1 + \lambda v_2$
10. *Distributivity of scalar multiplication over field addition:* $(\lambda + \mu)v = \lambda v + \mu v$

Remark 5. *We often use the term abstract vector space to emphasize that the elements of V are not restricted to arrows*[8] *or tuples*[9] *in $\mathbb{R}^n$, but can be any objects (like functions or tensors) that satisfy the axioms.*

3.1 Vectors

3.1.1 Vector operations

Vector operations are defined as follows:

- Addition: $\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{pmatrix}$.
- Scalar multiplication: $\lambda \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} \lambda x_1 \\ \vdots \\ \lambda x_n \end{pmatrix}$, where $\lambda \in \mathbb{R}$.

Definition 18 (Linear combination)**.** *A linear combination of vectors $v_1, \ldots, v_n$ is a vector v of the form $v = \lambda_1 v_1 + \ldots + \lambda_n v_n$, where $\lambda_1, \ldots, \lambda_n \in \mathbb{R}$.*

3.2 Span

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Definition 19 (Span)**.** *Let $A \subset \mathbb{R}^n$. The span of A, denoted* $\operatorname{span}(A)$*, is the set of all linear combinations of elements of A. In particular, if $A = \{v_1, \ldots, v_k\}$ (where k is the number of vectors in the set), then*

$$\operatorname{span}(A) = \{\lambda_1 v_1 + \cdots + \lambda_k v_k : \lambda_i \in \mathbb{R}\}.$$

[8] An arrow is a function or mapping between two sets.

[9] A tuple is an ordered sequence where duplicates are allowed, whereas a set is an unordered collection of unique elements.

When working in $\mathbb{R}^n$, the span describes all points you can reach by scaling and adding the given vectors. Depending on the vectors, the span can be a line (if the vectors are dependent), a plane, or a higher-dimensional subspace.

Remark 6. *If $A = \{v\} \subset \mathbb{R}^2$ with $v \neq 0$, then*

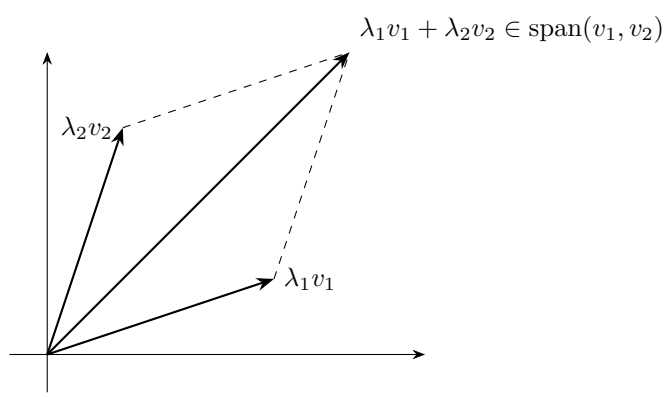
$$\operatorname{span}(v) = \{\lambda v : \lambda \in \mathbb{R}\}.$$

This set consists of all scalar multiples of v, which form a line in the direction of v. In particular, taking $\lambda = 0$ gives $0 \in \operatorname{span}(v)$, so the line passes through the origin.

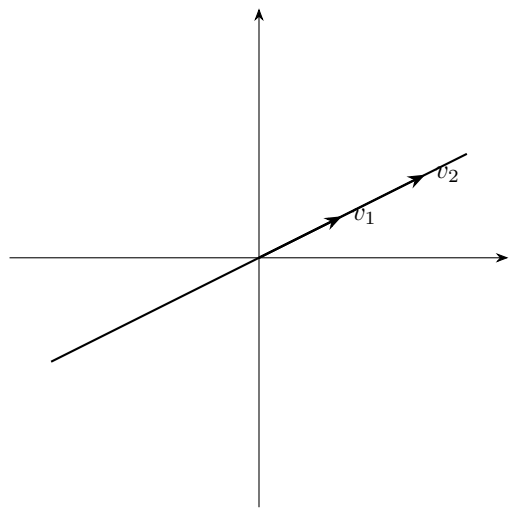
Example. *Let $A = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Then $\operatorname{span}(A) = \left\{ t \begin{pmatrix} 1 \\ 1 \end{pmatrix} \mid t \in \mathbb{R} \right\}$, which is a line in $\mathbb{R}^2$.*

Problem 4 (Span)**.** *Let $v_1 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$, $v_2 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$. Find $\operatorname{span}(v_1, v_2)$.*

Span is a generalization of lines in $\mathbb{R}^2$. For example, let v_1 and v_2 be two linearly independent vectors in $\mathbb{R}^2$. Any vector in $\operatorname{span}(v_1, v_2)$ has the form $\lambda_1 v_1 + \lambda_2 v_2$. Geometrically, this can be illustrated as the sum of two scaled vectors.



Furthermore, if v_1, v_2 are linearly dependent such that $v_1 = \lambda v_2$, where $\lambda \in \mathbb{R}$ and $\lambda \neq 0$, then $\operatorname{span}(v_1, v_2)$ is a line in $\mathbb{R}^2$ and $span(v_1) = span(v_2)$. Geometrically, this is a straight line through the origin in the direction of v_1 (and v_2).



Problem 5 (Span 2)**.** *Determine whether or not the first vector is in the span of the others. If so, write it as a linear combination of the other.*

$u = (2, 10, 7, 0), u_1 = (3, 10, 7, 0), u_2 = (1, 3, -2, 0), u_3 = (2, 8, 1, 0); \quad u, u_1, u_2, u_3 \in \mathbb{R}^4.$

3.2.1 Span in $\mathbb{C}^n$

Definition 20 (Span in $\mathbb{C}^n$)**.** *The span over $\mathbb{C}$ of vectors $v_1, \ldots, v_n \in \mathbb{C}^n$ is*

$$\operatorname{span}_{\mathbb{C}}(v_1, \ldots, v_n) = \{\lambda_1 v_1 + \cdots + \lambda_n v_n : \lambda_i \in \mathbb{C}\}.$$

Example. *Let*

$$v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 0 \\ i \end{pmatrix} \in \mathbb{C}^2.$$

Find $\operatorname{span}_{\mathbb{C}}(v_1, v_2)$*. For* $\lambda_1, \lambda_2 \in \mathbb{C}$*,*

$$\lambda_1 v_1 + \lambda_2 v_2 = \lambda_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} 0 \\ i \end{pmatrix} = \begin{pmatrix} \lambda_1 \\ i\lambda_2 \end{pmatrix}.$$

Since λ_1 is arbitrary and $i\lambda_2$ can represent any complex number, every vector in $\mathbb{C}^2$ can be written in this form. Hence

$$\operatorname{span}_{\mathbb{C}}(v_1, v_2) = \mathbb{C}^2.$$

3.3 Standard Basis

Definition 21 (Standard basis of $\mathbb{R}^n$)**.** *The standard basis of $\mathbb{R}^n$ is the set $\{e_1, \ldots, e_n\}$, where each vector e_i has a 1 in the i-th coordinate and 0 in all other coordinates:*

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \; e_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \; \ldots, \; e_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}.$$

Equivalently, the vectors $e_1, \ldots, e_n$ are the columns of the $n \times n$ identity matrix I_n.

Proposition 1. *Every vector* $x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n$ *can be written uniquely as* $x = x_1 e_1 + \cdots + x_n e_n$.

Remark 7. *To verify that the standard basis of $\mathbb{R}^n$ is indeed a basis of $\mathbb{R}^n$, we must check two properties: it spans $\mathbb{R}^n$ and it is linearly independent.*[10]

The standard basis also helps clarify the difference between real and complex vector spaces. In particular, the same vectors can generate very different spans depending on whether the scalars are real or complex.

Example (Real vs complex span of the same vectors)**.** *Let*

$$e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Over $\mathbb{C}$***:*** *Using complex scalars, any vector in $\mathbb{C}^2$ can be written as*

$$\lambda_1 e_1 + \lambda_2 e_2 = \begin{pmatrix} \lambda_1 \\ \lambda_2 \end{pmatrix}, \quad \lambda_1, \lambda_2 \in \mathbb{C}.$$

Thus $\operatorname{span}_{\mathbb{C}}(e_1, e_2) = \mathbb{C}^2$*, and only 2 vectors are needed:*

$$\dim_{\mathbb{C}} \mathbb{C}^2 = 2.$$

Over $\mathbb{R}$***:*** *Now only real scalars are allowed. Then*

$$a_1 e_1 + a_2 e_2 = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}, \quad a_1, a_2 \in \mathbb{R}.$$

Thus $\operatorname{span}_{\mathbb{R}}(e_1, e_2) = \mathbb{R}^2$*, and only 2 vectors are needed:* $\dim_{\mathbb{R}} \mathbb{R}^2 = 2$*.*
However, the standard basis of $\mathbb{R}^2$ is not enough to span $\mathbb{C}^2$, since it cannot produce vectors with imaginary parts. To span all of $\mathbb{C}^2$ over $\mathbb{R}$, we also need

$$ie_1 = \begin{pmatrix} i \\ 0 \end{pmatrix}, \quad ie_2 = \begin{pmatrix} 0 \\ i \end{pmatrix}.$$

Now any vector in $\mathbb{C}^2$ can be expressed as a real linear combination of the 4 vectors

$$ae_1 + be_2 + c(ie_1) + d(ie_2) = \begin{pmatrix} a + ci \\ b + di \end{pmatrix}, \quad a, b, c, d \in \mathbb{R}.$$

Thus $\operatorname{span}_{\mathbb{R}}(e_1, e_2, ie_1, ie_2) = \mathbb{C}^2$ *and therefore*

$$\dim_{\mathbb{R}} \mathbb{C}^2 = 4.$$

3.4 Coordinate Spaces

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Definition 22 (Coordinate Spaces)**.** *A coordinate space of dimension n over a field k ($k = \mathbb{R}$ or $k = \mathbb{C}$) is defined as*

$$k^n = \left\{ \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \mid x_i \in k \right\}.$$

The standard basis for a coordinate space is $\{e_1, \ldots, e_n\}$, where

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \ \ldots \ , e_n = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}.$$

The dimension of k^n is $\dim k^n = n$.

3.5 Polynomial Spaces

Definition 23 (Polynomial Spaces)**.** *A polynomial space of degree at most n over a field k is*

$$P_n(k) = \{a_n x^n + \cdots + a_1 x + a_0 \mid a_i \in k\},$$

which forms a vector space over k with standard basis $\{1, x, x^2, \ldots, x^n\}$. Under the identification $P_n(k) \cong k^{n+1}$, these basis elements have coordinate vectors

$$1 \leftrightarrow \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad x \leftrightarrow \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \quad \ldots, \quad x^n \leftrightarrow \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}.$$

The dimension of $P_n(k)$ is $\dim P_n(k) = n + 1$.

Example. *Examples of polynomials in these spaces are*

$$1 + x^2 \in P_2(\mathbb{R}), \qquad 1 + ix^3 \in P_3(\mathbb{C}).$$

In each case the degree of the polynomial satisfies $\deg(f) \leq n$.

Remark 8. *All polynomials over k form the space*

$$P_\infty(k) = \{a_n x^n + \cdots + a_1 x + a_0 : a_i \in k, \ n \geq 0\},$$

which is an infinite-dimensional vector space over k.

Remark 9. *The polynomial spaces form a nested sequence*

$$P_0(k) \subseteq P_1(k) \subseteq P_2(k) \subseteq \cdots \subseteq P_\infty(k).$$

[10] To prove the standard basis is a basis, please refer to the Appendix (6.1).

3.6 Matrix Spaces

Definition 24 (Matrix Spaces)**.** *The set of all $n \times n$ matrices over a field k is*

$$M_n(k) = \left\{ \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \mid a_{ij} \in k \right\},$$

which forms a vector space over k. A standard basis for $M_n(k)$ is the set of matrices $\{e_{ij} : 1 \leq i, j \leq n\}$, where e_{ij} has a 1 in the (i, j)-th entry and 0 elsewhere. For example,

$$e_{11} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}, \quad e_{12} = \begin{pmatrix} 0 & 1 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}, \quad e_{21} = \begin{pmatrix} 0 & 0 & \cdots & 0 \\ 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}.$$

The dimension of $M_n(k)$ is $\dim M_n(k) = n^2$.

Example (Matrix Spaces)**.** *For $M_2(\mathbb{R})$,*

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = 1 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + 2 \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + 3 \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + 4 \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = 1e_{11} + 2e_{12} + 3e_{21} + 4e_{22}.$$

3.7 Function Spaces

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Definition 25 (Function spaces)**.** *Let D be a set and k a field. Define*

$$F(D, k) = \{f : D \to k\}.$$

With pointwise operations

$$(f + g)(x) = f(x) + g(x), \quad (\lambda f)(x) = \lambda f(x),$$

$F(D, k)$ is a vector space over k.

A function $f \in F(D, k)$ is completely determined by its values on D. If $D = \{1, \ldots, n\}$, this gives n numbers $f(1), \ldots, f(n)$. To construct a basis, we introduce functions that isolate each of these values.

1. Finite case Assume $D = \{1, \ldots, n\}$. For each $i \in D$, define the Kronecker delta function

$$\delta_i(j) = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Then $\{\delta_1, \ldots, \delta_n\}$ is a basis of $F(D, k)$. Moreover, every $f \in F(D, k)$ can be written uniquely as

$$f = \sum_{i=1}^{n} f(i)\, \delta_i.$$

Indeed, evaluating at j gives

$$\sum_{i=1}^{n} f(i)\delta_i(j) = f(j),$$

since $\delta_i(j) = 0$ unless $i = j$. Hence the functions span, and linear independence follows by evaluating a linear relation at each point of D. Therefore

$$\dim F(D, k) = n.$$

2. Infinite case If D is infinite, $F(D, k)$ has no finite basis. The family $\{\delta_x : x \in D\}$ is linearly independent but does not span $F(D, k)$.

Example (Vector space of functions)**.** *Let $D = \{1, 2, 3\}$ and $k = \mathbb{R}$. Consider the vector space $F(D, \mathbb{R}) = \{f : D \to \mathbb{R}\}$. Find a basis of $F(D, \mathbb{R})$ and express $f(1) = 2, f(2) = -1, f(3) = 3$ as a linear combination of the basis functions.*
1. Define the standard (Kronecker delta) functions:

$$\delta_1(j) = \begin{cases} 1, & j = 1 \\ 0, & j \neq 1 \end{cases}, \quad \delta_2(j) = \begin{cases} 1, & j = 2 \\ 0, & j \neq 2 \end{cases}, \quad \delta_3(j) = \begin{cases} 1, & j = 3 \\ 0, & j \neq 3 \end{cases}.$$

Then $\{\delta_1, \delta_2, \delta_3\}$ is a basis of $F(D, \mathbb{R})$.
2. Write f as a linear combination:

$$f = f(1)\delta_1 + f(2)\delta_2 + f(3)\delta_3 = 2\delta_1 - 1\delta_2 + 3\delta_3.$$

3. Evaluating at each point:

$$f(1) = 2 \cdot 1 + (-1) \cdot 0 + 3 \cdot 0 = 2, \quad f(2) = 2 \cdot 0 + (-1) \cdot 1 + 3 \cdot 0 = -1,$$

$$f(3) = 2 \cdot 0 + (-1) \cdot 0 + 3 \cdot 1 = 3.$$

Hence the decomposition is correct.

3.8 Subspaces

January 19, 2026

Definition 26 (Subspace)**.** *A subspace is a subset of a larger vector space that is itself a vector space. A subspace of V is a subset U of V which is a vector space (over the same field k) with respect to the operations of addition and scalar multiplication induced from V.*

Proposition 2 (Subspace criterion)**.** *Let V be a vector space over a field k and let $U \subseteq V$. Then U is a vector subspace of V if and only if:*

1. $0 \in U$,
2. $u, v \in U \Rightarrow u + v \in U$,
3. $u \in U,\ \lambda \in k \Rightarrow \lambda u \in U$.

Problem 6 (Subspace criterion)**.** *Let $A \in M_n(k)$ be a fixed matrix. Prove that*

$$U = \{Ax = \vec{0} \mid x \in k^n\}$$

is a subspace, null space or kernel.

Problem 7 (Subspace criterion 2)**.** *Let $A \in M_n(k)$. Show that*

$$U = \{Ax \mid x \in K^n\}$$

is a subspace of k^n. The set U is called the image (or range) of A.

Problem 8 (Subspace criterion 3)**.** *Let $V = k$, viewed as a vector space over k. Show that the only subspaces of V are $\{0\}$ and V itself.*

Problem 9 (Subspace criterion 4)**.** *Let V be a vector space over k and let $v_1, \ldots, v_n \in V$. Show that*

$$\operatorname{span}(v_1, \ldots, v_n) = \{\lambda_1 v_1 + \cdots + \lambda_n v_n \mid \lambda_1, \ldots, \lambda_n \in k\}$$

is a subspace of V.

January 21, 2026

3.8.1 Membership in a span

Problem 10 (Span membership)**.** *Let*

$$v = \begin{pmatrix} 3 \\ 5 \\ -5 \end{pmatrix}, \quad v_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \; v_2 = \begin{pmatrix} 2 \\ 5 \\ 4 \end{pmatrix}, \; v_3 = \begin{pmatrix} 1 \\ 3 \\ 6 \end{pmatrix}.$$

Decide whether $v \in \operatorname{span}(v_1, v_2, v_3)$.

Problem 11 (Span membership 2)**.** *Let*

$$f = 3x^2 + 5x - 5, \quad f_1 = x^2 + 2x + 1, \; f_2 = 2x^2 + 5x + 4, \; f_3 = x^2 + 3x + 6.$$

Decide whether $f \in \operatorname{span}(f_1, f_2, f_3)$.

3.8.2 Operations on subspaces

Let W be a vector space and $U, V \subseteq W$ subspaces.

Definition 27 (Intersection)**.**

$$U \cap V = \{w \in W \mid w \in U \text{ and } w \in V\}.$$

The intersection of subspaces is a subspace. In particular, the smallest possible intersection is $\{0\}$.

Definition 28 (Union)**.**

$$U \cup V = \{w \in W \mid w \in U \text{ or } w \in V\}.$$

The union of subspaces is not necessarily a subspace.

Proposition 3. *In general, $U \cup V$ is a subspace of W only if one subspace is contained in the other, i.e., $U \subseteq V$ or $V \subseteq U$.*

Proof. First assume $U \subseteq V$. Then

$$U \cup V = V,$$

which is a subspace of W. Similarly, if $V \subseteq U$, then

$$U \cup V = U,$$

which is also a subspace. Conversely, suppose neither $U \subseteq V$ nor $V \subseteq U$. Then there exist $u \in U$ with $u \notin V$, and $v \in V$ with $v \notin U$. Thus $u, v \in U \cup V$. If $U \cup V$ were a subspace, it would be closed under addition, so

$$u + v \in U \cup V.$$

If $u + v \in U$, then since U is a subspace and $u \in U$,

$$v = (u + v) - u \in U,$$

a contradiction. If $u + v \in V$, then since V is a subspace and $v \in V$,

$$u = (u + v) - v \in V,$$

also a contradiction. Therefore $u + v \notin U \cup V$, so $U \cup V$ is not closed under addition. Hence $U \cup V$ is not a subspace if neither $U \subseteq V$ nor $V \subseteq U$. $\square$

Definition 29 (Sum of subspaces)**.**

$$U + V = \{u + v \mid u \in U,\, v \in V\}.$$

Proposition 4. *If U and V are subspaces of W, then $U + V$ is a subspace of W.*

Proof. We apply the subspace criterion.

1. Since $0 \in U$ and $0 \in V$, we have $0 = 0 + 0 \in U + V$.
2. Let $w_1, w_2 \in U + V$. Then

   $$w_1 = u_1 + v_1, \quad w_2 = u_2 + v_2,$$

   with $u_1, u_2 \in U$ and $v_1, v_2 \in V$. Hence

   $$w_1 + w_2 = (u_1 + u_2) + (v_1 + v_2) \in U + V.$$

3. Let $w \in U + V$ and $\lambda \in K$. Then $w = u + v$ and

$$\lambda w = \lambda u + \lambda v \in U + V.$$

□

Proposition 5.

$$U + V = \operatorname{span}(U \cup V).$$

Equivalently, $U + V$ is the smallest subspace of W containing both U and V.

Proof. ($\subseteq$): Let $w \in U + V$. Then $w = u + v$ with $u \in U$, $v \in V$. Since $u, v \in U \cup V$, we have $w \in \operatorname{span}(U \cup V)$.

($\supseteq$): Let $w \in \operatorname{span}(U \cup V)$. Then

$$w = a_1 u_1 + \cdots + a_n u_n + b_1 v_1 + \cdots + b_m v_m$$

with $u_i \in U$, $v_j \in V$. Grouping terms,

$$w = (a_1 u_1 + \cdots + a_n u_n) + (b_1 v_1 + \cdots + b_m v_m),$$

so $w \in U + V$. □

January 23, 2026

3.9 Direct sums

Let W be a vector space and $U, V \subseteq W$ be subspaces.

Definition 30 (Direct sum)**.** *We say that U and V are in direct sum if*

$$U \cap V = \{0\}.$$

In this case, their sum

$$U + V = \{u + v \mid u \in U,\ v \in V\}$$

is denoted by $U \oplus V$. We write

$$W = U \oplus V \iff \begin{cases} W = U + V, \\ U \cap V = \{0\}. \end{cases}$$

Proof. Assume $W = U \oplus V$. Then $W = U + V$ and $U \cap V = \{0\}$. First, since $W = U + V$, every $w \in W$ can be written as

$$w = u + v, \quad u \in U,\ v \in V.$$

To show uniqueness, suppose

$$w = u_1 + v_1 = u_2 + v_2$$

with $u_1, u_2 \in U$ and $v_1, v_2 \in V$. Then

$$u_1 - u_2 = v_2 - v_1.$$

Since U is a subspace, $u_1 - u_2 \in U$. Since V is a subspace, $v_2 - v_1 \in V$. Hence

$$u_1 - u_2 \in U \cap V.$$

But $U \cap V = \{0\}$, so

$$u_1 - u_2 = 0.$$

Thus $u_1 = u_2$, and therefore $v_1 = v_2$. Hence the decomposition is unique. Conversely, assume every $w \in W$ can be written uniquely as $w = u + v$ with $u \in U$, $v \in V$. Then clearly $W = U + V$. To show $U \cap V = \{0\}$, let $w \in U \cap V$. Then

$$w = w + 0 = 0 + w,$$

where $w \in U$, $0 \in V$, and $0 \in U$, $w \in V$. By uniqueness of the decomposition, we must have

$$w = 0.$$

Thus $U \cap V = \{0\}$, so $W = U \oplus V$. □

3.9.1 Analogy with sets

Sets	Vector spaces
$A \cap B$	$U \cap V$
$A \cup B$	$U + V$
$A \sqcup B,\ A \cap B = \emptyset$	$U \oplus V,\ U \cap V = \{0\}$

Example (Analogy with disjoint sets)**.** *Let*

$$D = \{1, 2, 3, 4, 5\}, \quad A = \{1, 2, 3\}, \quad B = \{4, 5\}.$$

Then

$$D = A \sqcup B.$$

3.9.2 Direct sums in function spaces

Definition 31 (Subspace of functions supported on a subset)**.** *Let D be a set and $A \subseteq D$. Consider the function space $F(D, \mathbb{R})$. Define*

$$U = \{f \in F(D, \mathbb{R}) : f(x) = 0 \, \forall x \notin A\}.$$

Then U is a subspace of $F(D, \mathbb{R})$ (it is closed under addition and scalar multiplication, and contains the zero function). Moreover, U can be naturally identified with $F(A, \mathbb{R})$, since each function in U is completely determined by its values on A.

Remark 10. *Every function in U is completely determined by its values on A and vanishes outside A.*

Example (Functions supported on a subset)**.** *Let* $D = \{1, 2, 3, 4, 5\}$ *and* $A = \{2, 4\}$*. Then*

$$U = \{f \in F(D, \mathbb{R}) : f(1) = f(3) = f(5) = 0\}.$$

Examples of functions in U *include*

$$f(1) = 0, f(2) = 3, f(3) = 0, f(4) = -1, f(5) = 0,$$

$$g(1) = 0, g(2) = 0, f(3) = 0, g(4) = 7, g(5) = 0.$$

Each function in U *is completely determined by its values on* A*, i.e.,*

$$f \iff (f(2), f(4)) \in F(A, \mathbb{R}) \cong \mathbb{R}^2.$$

Hence U *is a 2-dimensional subspace of* $F(D, \mathbb{R})$*.*

4 Basis and Dimension

January 26, 2026

4.1 Finite Dimensional Spaces

Let V be a vector space over a field k. Then there are two possibilities:

- The zero vector space: $V = \{0\}$
- The non-zero vector space: $V \neq \{0\}$

In the second case, there exists a non-zero vector $v_1 \in V$ such that $v_1 \neq 0$. Then $span(v_1)$ is a subspace of V, and there are two possibilities:

- $V = span(v_1)$, i.e., v_1 is a generator of V.
- $V \neq span(v_1)$, i.e., there exists a $v_2 \in V$ such that $v_2 \notin span(v_1)$.

In the second case, $span(v_1, v_2) \subseteq V$. This process can be repeated to obtain a sequence of vectors $v_1, v_2, \ldots, v_n$ such that $span(v_1, \ldots, v_n) \subseteq V$. The maximum number of linearly independent vectors in V is called the dimension of V, denoted by $\dim V$.

2 cases (2d space plane):

- If $V = span(v_1, v_2)$, you have a 2D plane.
- If the space is larger than that plane, there must exist a v_3 that sticks out of it, meaning $span(v_1, v_2, v_3) \subseteq V$.

2 cases (3d space):

- If v_3 covers the whole space ($v_3 = V$), you are in 3D.
- If not, there is a v_4, and so on.

Definition 32 (Finite Dimensional Vector Space)**.** *A vector space V over a field k is finite-dimensional if there exists a finite set of vectors $v_1, \dots, v_n \in V$ such that*

$$V = \operatorname{span}(v_1, \dots, v_n).$$

The dimension of V is the minimal such n:

$$\dim_k V = \min\{n : V = \operatorname{span}_k(v_1, \dots, v_n)\}.$$

Example (Complex numbers as a real vector space)**.** *Consider $\mathbb{C}$ as a vector space over $\mathbb{R}$. A standard basis is*

$$\{1, i\},$$

so $\dim_{\mathbb{R}} \mathbb{C} = 2$. Any $z \in \mathbb{C}$ can be written uniquely as

$$z = x \cdot 1 + y \cdot i, \quad x, y \in \mathbb{R}.$$

Now consider the space of $n \times n$ complex matrices $M_n(\mathbb{C})$. Each entry $z_{ij} \in \mathbb{C}$ can be written as $z_{ij} = x_{ij} + y_{ij} i$ with $x_{ij}, y_{ij} \in \mathbb{R}$. Hence a standard basis over $\mathbb{C}$ consists of the matrices E_{ij} with a 1 in the (i, j)-th entry and 0 elsewhere, giving

$$\dim_{\mathbb{C}} M_n(\mathbb{C}) = n^2.$$

Over $\mathbb{R}$, each E_{ij} splits into two real matrices corresponding to the real and imaginary parts, so

$$\dim_{\mathbb{R}} M_n(\mathbb{C}) = 2n^2.$$

Example (Infinite-dimensional space of polynomials)**.** *Let*

$$P_\infty(\mathbb{R}) = \{a_0 + a_1 x + a_2 x^2 + \cdots : a_i \in \mathbb{R}\}$$

be the space of all polynomials with real coefficients.

Define the sequence of vectors

$$v_1 = 1, \quad v_2 = x, \quad v_3 = x^2, \quad \dots$$

Then

$$\textit{span}\{v_1\} = \textit{constant polynomials}, \quad \textit{span}\{v_1, v_2\} = P_1,$$

and in general

$$v_{n+1} = x^n \notin P_{n-1} = \textit{span}\{v_1, \dots, v_n\}.$$

*Thus $P_\infty(\mathbb{R})$ is **infinite-dimensional**, and a standard basis is*

$$\{1, x, x^2, x^3, \dots\}.$$

4.2 Linear Independence

Definition 33. *$v_1, \ldots, v_n$ are linearly independent if $\lambda_1 v_1 + \ldots + \lambda_n v_n = 0 \implies \lambda_1 = \ldots = \lambda_n = 0$.*

Example.

$$e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Prove LI. Assume $xe_1 + ye_2 = 0$. Prove $x = y = 0$.

$$xe_1 + ye_2 = x \begin{pmatrix} 1 \\ 0 \end{pmatrix} + y \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} = 0 \implies x = y = 0.$$

Proposition 6. *The "standard basis" are both spanning and LI. $(e_1, \ldots, e_n)$ of k^n $(1, x, \ldots, x^n)$ of P_n $(e_{11}, \ldots, e_{nn})$ of $M_n(k)$ $(\delta_1, \ldots, \delta_n)$ of $F(D)$, where $D = \{1, \ldots, n\}$*

Definition 34. *basis $= set(v_1, \ldots, v_n)$ both span and LI.*

$$dim(\forall) = \#basis$$

Problem 12. *Prove $dimP_\infty = \infty$. Assume $dimP_\infty < \infty$. Proof by contradiction. $\exists f_1, \ldots, f_n$ polynomials $P_\infty = span(f_1, \ldots, f_n)$. Let $d =$ max degree of f_k.*

$$x^{d+1} \notin span(f_1, \ldots, f_n)$$

degree $\leq d$.

January 28, 2026

Problem 13 (Linearly independent vectors)**.**

$$v_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 4 \\ 9 \\ 5 \end{pmatrix}.$$

Are these vectors linearly independent? Are there $x, y, z \in \mathbb{R}$ such that

$$xv_1 + yv_2 + zv_3 = 0?$$

Definition 35 (Linear Dependence)**.** *Linearly dependent (LD) just means not LI.*

Proposition 7. *$v_1, \ldots, v_n$ are linearly dependent if and only if one vector is in the span of the other vectors.*

Example.

$$3v_1 + 5v_2 - 2v_3 = 0$$

As we can see, v_1 is in the span of v_2 and v_3.

$$v_1 = \frac{1}{3}(-5v_2 + 2v_3)$$

Remark 11. *If one vector $v_1 = 0$ then the family is LD.*

Proof. Proof of $\longrightarrow$: Assumption $\lambda_1 v_1 + \cdots + \lambda_n v_n = 0$. Not all λ_i are zero, at least one is nonzero, say $\lambda_k \neq 0$. The goal is to write v_k in the span of the other vectors.

$$v_k = \frac{-1}{\lambda_k} \sum_{i \neq k} \lambda_i v_i.$$

$$v_k \in span(v_1, \ldots, v_k, \ldots, v_n).$$

Converse: Assume one vector, say v_k is in the span of the other vectors.

$$v_k \in span(v_1, \ldots, v_k, \ldots, v_n).$$

There exists coefficients $\lambda_1, \ldots, \lambda_n$ such that

$$v_k = \lambda_1 v_1 + \cdots + \lambda_{k-1} v_{k+-1} + \lambda_{k+1} v_{k+1} + \cdots + \lambda_n v_n.$$

LD: non trivial dependency between the v_i's. □

Example.

$$V = \mathbb{R}^3$$

$$v_1 = \begin{pmatrix} a_1 \\ 0 \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} b_1 \\ b_2 \\ 0 \end{pmatrix}, \quad v_3 = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}.$$

LI:

$$v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

LD:

$$v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix}.$$

Assume $xv_1 + yv_2 + zv_3 = 0$.

$$x \begin{pmatrix} a_1 \\ 0 \\ 0 \end{pmatrix} + y \begin{pmatrix} b_1 \\ b_2 \\ 0 \end{pmatrix} + z \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}.$$

$$\begin{cases} a_1 x + b_1 y + c_1 z = 0 \\ b_2 y + c_2 z = 0 \\ c_3 z = 0 \end{cases}$$

If $a_1 = 0$. $v_1 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$. Answer: LD. Can assume $a_1 \neq 0$. If $b_2 = 0$. $v_1 = \begin{pmatrix} a_1 \\ 0 \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} b_1 \\ 0 \\ 0 \end{pmatrix}$. *Answer: LD.*

January 30, 2026

4.3 Basis

Definition 36 (Basis)**.** *Let V be a vector space. A set of vectors $B = \{v_1, \dots, v_k\} \subset V$ is called a basis of V if*

1. *B spans V, and*
2. *B is linearly independent.*

Equivalently, every vector in V can be written uniquely as a linear combination of the vectors in B.

Example (Standard basis of $\mathbb{R}^3$)**.**

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

These vectors form a basis of $\mathbb{R}^3$.

Proof. Let $v = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$. Then

$$v = xe_1 + ye_2 + ze_3$$

so the set spans $\mathbb{R}^3$. If

$$xe_1 + ye_2 + ze_3 = 0 \implies x = y = z = 0,$$

hence the vectors are linearly independent. □

Proposition 8 (Common standard bases)**.**

- $\mathbb{R}^n$*:* $(e_1, \dots, e_n)$
- $M_n(k)$*:* e_{ij}
- $P_n(k)$*:* $(1, x, \dots, x^n)$
- $F(D)$ *for finite* D*:* $\{\delta_x, x \in D\}$

4.3.1 Size of spanning and independent sets

For $(v_1, \dots, v_n) \in \mathbb{R}^3$:

- If $n < 3$ then the vectors cannot span $\mathbb{R}^3$.
- If $n > 3$ then the vectors cannot be linearly independent.

Lemma 2. *If $u_1, \dots, u_n$ span V and $v_1, \dots, v_m$ are linearly independent in V, then*

$$n \geq m.$$

We can conclude, that a spanning set $\geq$ LI set.

4.3.2 Characterizations of a basis

Theorem 3 (Equivalent Conditions)**.** *Let $(v_1, \ldots, v_n)$ be vectors in V. The following are equivalent:*

1. *Basis: $(v_1, \ldots, v_n)$ is a basis.*
2. *Minimal span: It spans V and n is minimal among spanning sets.*
3. *Maximal LI: It is linearly independent and n is maximal among independent sets.*
4. *Coordinates: Every $v \in V$ has a unique representation.*

$$v = x_1 v_1 + \cdots + x_n v_n.$$

The vector

$$[v]_e = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

is called the coordinate vector of v.

Corollary 3.1. *Any two bases of V have the same number of vectors. This number is the dimension of V.*

Proof. Let $(e_1, \ldots, e_n)$ and $(f_1, \ldots, f_m)$ be bases. Since one spans and the other is linearly independent, the lemma gives $n \geq m$ and $m \geq n$, hence $n = m$. □

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4.3.3 Finding a Basis

Let $(v_1, \ldots, v_n) \in k^m$ and set $U = \operatorname{span}(v_1, \ldots, v_n)$.

1. Put the vectors as rows of a matrix A.
2. Row reduce A to echelon form B.
3. The nonzero rows of B form a basis of U.

1. Proving a set of vectors is a Basis Let V be finite dimensional.
Method 1 (dimension test).

$$(v_1, \ldots, v_n) \text{ is a basis} \iff \dim(\operatorname{span}(v_1, \ldots, v_n)) = n.$$

Compute the dimension using row reduction.
Method 2 (basis criterion when $\dim V = n$).

Proposition 9. *Let $\dim V = n$ and $v_1, \ldots, v_n \in V$. The following are equivalent:*

1. $(v_1, \ldots, v_n)$ *is a basis,*

2. $(v_1, \ldots, v_n)$ *spans* V*,*

3. $(v_1, \ldots, v_n)$ *is linearly independent.*

Idea. A spanning set of $n = \dim V$ vectors is minimal, hence linearly independent. An independent set of n vectors is maximal, hence spans. □

2. Transfer (exchange) principle If

$$u_1, \ldots, u_n \text{ span } V \quad \text{and} \quad v_1, \ldots, v_n \text{ are linearly independent,}$$

then both lists are bases of V.

Example. *In* P_n*,*

$$(1, x, \ldots, x^n) \text{ spans}, \qquad ((x-a)^0, \ldots, (x-a)^n) \text{ is LI},$$

so both are bases.

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4.3.4 Direct Sums and Unique Decomposition

Let W be a vector space and $U, V \subseteq W$ subspaces.

Definition 37 (Direct Sum)**.** *The sum of subspaces is*

$$U + V = \{u + v : u \in U,\, v \in V\}.$$

We say the sum is direct, written $U \oplus V$*, if*

$$U \cap V = \{0\}.$$

Proposition 10. *The following are equivalent:*

1. $U \oplus V$ *(i.e.* $U \cap V = \{0\}$*),*

2. *every* $w \in U + V$ *can be written uniquely as* $w = u + v$ *with* $u \in U$*,* $v \in V$*.*

Proof. ($\Rightarrow$) Suppose $w = u_1 + v_1 = u_2 + v_2$. Then

$$u_1 - u_2 = v_2 - v_1 \in U \cap V = \{0\},$$

so $u_1 = u_2$ and $v_1 = v_2$.

($\Leftarrow$) If decomposition is unique and $w \in U \cap V$, then

$$0 = 0 + 0 = w + (-w),$$

so uniqueness forces $w = 0$. □

Coordinates relative to a basis If $(e_1, \ldots, e_n)$ is a basis of V, every $v \in V$ can be written uniquely as

$$v = \lambda_1 e_1 + \cdots + \lambda_n e_n.$$

The vector

$$[v]_e = \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix}$$

is called the coordinate vector of v.

Example. *In $\mathbb{R}^3$ with the standard basis,*

$$v = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad \Rightarrow \quad [v]_e = \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

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example:

$$v = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, f_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, f_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

$$[v]_f = \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}$$

LL example: Find coordinate of $v = \begin{pmatrix} 5 \\ 3 \\ 4 \end{pmatrix}$ in basis $f = (f_1, f_2, f_3)$, with $f_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, f_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, f_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$.

$$\begin{pmatrix} 5 \\ 3 \\ 4 \end{pmatrix} = x \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + y \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + z \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

Find x, y, z.

$$[v]_f = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 4 \end{pmatrix} \in \mathbb{R}^3.$$

Example: $D = \{0, 1, 2, 3, 4\}$, $V = F(D)$, $g(x) = x^2$, $g \in V$, find coordinate of y in standard basis $\delta = (\delta_0, \delta_1, \delta_2, \delta_3, \delta_4)$. $dimV = 5$. $[g]_\delta = \begin{pmatrix} \lambda_0 \\ \vdots \\ \lambda_4 \end{pmatrix}$, where $g = \lambda_0\delta_0 + \lambda_1\delta_1 + \lambda_2\delta_2 + \lambda_3\delta_3 + \lambda_4\delta_4$.

Nose Basis theorem: Basis extraction theorem: $v_1, \ldots, v_n$ spans V can extract a basis $\{e_1, \ldots, e_n\}$ from $\{v_1, \ldots, v_n\}$. Example: $v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, v_2 =$

$\begin{pmatrix}0\\1\end{pmatrix}, v_3 = \begin{pmatrix}1\\1\end{pmatrix}$ spans $\mathbb{R}^3$. Proof: Let $\{e_1, \ldots, e_n\} \subseteq \{v_1, \ldots, v_n\}$ be a subset which is spanning with m minimal. Claim: $e_1, \ldots, e_m$ is a basis. Proof by contradiction: Asumption: $(e_1, \ldots, e_n)$ not LI: Proposition: one vector is in the span of the other $(\lambda_1 e_1 + \ldots + \lambda_n e_n = 0)$. $e_k \in span(e_1, e_2, \ldots, e_k, \ldots, e_n)$.

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Vvector space (finite dimensional) $v_1, \ldots, v_n$ LI set. Can extend $v_1, \ldots, v_n$ to a basis of V. Proof: $span(v_1, \ldots, v_n) = U \subseteq V$. Step 1: Case 1: If $U = V \to$ basis (LI + span) Case 2: $U \neq V \implies \exists v_{n+1} \in V\ U$ Claim: $(v_1, \ldots, v_{n+1})$ LI V minus U.

$$U_{n+1} = span(v_1, \ldots, v_{n+1})$$

Step 2: Case 1: $U_{n+1} = V \implies (v_1, \ldots, v_{n+1})$ is a basis. Case 2: $U_{n+1} \neq V \implies$ a m add v_{m+2}

Step 3: ...This stops at v_n for some $n \geq m$, $n = dimV \implies n$ LI vectors in V.

Coronary: $U \subseteq V$, $dimU \leq dimV$ or $U \nsubseteq V$ and $dimU < dimV$.[11]. If for $U \subseteq V$, $U = V \iff dimU = dimV$.

Coronary: $U \subseteq V$. U admits a supplement, meaning there exists a subspace $U' \subseteq V$ such that $V = U \oplus U'$. $dimV = dimU + dimU'$.

Basis concatenation: Theorem: Suppose $U \oplus U'$ are in direct sum in V. $(e_1, \ldots, e_n)$ basis of U. $(f_1, \ldots, f_m)$ basis of U'. $\implies (e_1, \ldots, e_n, f_1, \ldots, f_m)$ is a basis of $U \oplus U'$, which means $dimU \oplus U' = m + n = dimU + dimU'$. Theorem: $U, U' \subset V$, $dimU + U' = dimU + dimU' - dim(U \cap U')$.

proof of : Suppose $U \oplus U'$ are in direct sum in V Assume: $\begin{cases} e_1, \ldots, e_n basisofU \\ f_1, \ldots, f_m basisofU' \end{cases}$

Prove $(e_1, \ldots, e_n, f_1, \ldots, f_m)$ is a basis LI: $a_1e_1 + \ldots + a_ne_n + b_1f_1 + \ldots + b_mf_m = 0$ $w = a_1e_1 + \ldots + a_ne_n = -b_1f_1 - \ldots - b_mf_m$. $w \in U$, $w \in U' \implies w \neq 0 \implies a_1e_1 + \ldots a_ne_n = 0 \implies a_i = 0. \implies b_1f_1 + \ldots b_nf_n = 0 \implies b_i = 0.$

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$Span : v \in U + U' \implies v = u + u'$

5 Linear Transformation

Today general trans/function/map U,V are sets where U is the domain and V is the codomain. $u \in U$ and $T(u) \in V$. Image of $T = Im(T) = \{T(u) : u \in U\} \subseteq V$. Example: $sin : \mathbb{R} \to \mathbb{R}$, $x \to sin(x)$. $Im(sin) = (-1, 1)$. $F(D)$ is a vector space with D set, codomain $K = \mathbb{R} or \mathbb{C}$. Definition: $T : U \to V$, $u \to T(u)$ is surjective, if $\Im(T) = V$. Definition: $T : U \to V$, $u \to T(u)$ is injective, if you have u and v where $u \neq v$, but $T(u) = T(v)$. Injective: if $u, v \in U$ satisfy $T(u) = T(v)$, then $u = v$. Example: x^3 is injective. Definition: T is a bijection if it is both injective and surjective. Example: x^3 is bijective whilst x^2 is not. More precisely: $f : \mathbb{R} \to \mathbb{R}\ x \to x^2$ is not injective nor surjective. Example: $y : \mathbb{C} \to \mathbb{C}\ z \to z^2$ injective? $(-1)^2 = (1)^2$. surjective? $z^2 = w$ where $w \in \mathbb{C}$ is any complex number. Fix w, solve for z, $g(z) = w$.

[11] $\nsubseteq$ means it is a subset but not equal to.

Composition of transformations:

$$R :\longrightarrow T \longrightarrow V \longrightarrow S \longrightarrow W$$

$$u \longrightarrow T(u) \longrightarrow S(T(u))$$

Composition:

$$R : U \longrightarrow W$$

$$u \longrightarrow S(T(u))$$

$$g : \mathbb{R} \xrightarrow[A]{x^2} \mathbb{R} \xrightarrow[B]{sin} \mathbb{R}$$

$$g = sin(x^2)$$

$$R : U \xrightarrow{T} V \xrightarrow{S} W$$

$$R(u) = S(T(u))$$

$$R = S \circ T$$

($\circ$) is called composition. A inverse = a matrix B such that $AB = Id$ and $BA = Id$. Proposition: $T : U \to V$ is bjective $\iff$ T admits an inverse, $\exists S$ such that $\begin{cases} ST = id \\ TS = id \end{cases}$. Let U be a vector space:

$$T : U \longrightarrow U \text{Id transformation}$$

$$u \longrightarrow u$$

$$U = \mathbb{R}^n$$

$$T \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

Example: $U = \mathbb{R}^n$, $V = \mathbb{R}^n$, $A \in M_n(\mathbb{R})$, $T_A : u \to v$, $T_A =$ trans with A.

$$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \to A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

Example:

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

$$A \begin{pmatrix} x \\ y \end{pmatrix}$$

$$T : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$$

$$\begin{pmatrix} x \\ y \end{pmatrix} \to \begin{pmatrix} x+2y \\ 3x+4y \end{pmatrix}$$

Definition: U,V vector spaces over K. $T : U \to V$ transformation. Say T is linear if $T(u+v) = T(u) + T(v)$ and $T(\lambda u) = \lambda T(u)$. Proposition: S,T are linear, here $S \circ T$ is linear.

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Linear transformations:

$$T : U \longrightarrow V$$
$$u \longrightarrow T(u)$$

Imagine two big circles one is U the othre V, in each a small dot $u \in U$ and $v \in V$ and an arrow from u to v, and this arrow is T. T linear:

1. Additive: $T(u+u') = T(u) + T(u')$
2. Scalars: $T(\lambda u) = \lambda T(u)$

Main example: $T : \mathbb{R}^n \to \mathbb{R}^n$ (a matrix).

$$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \to A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

Example 1:

$$T : \mathbb{R}^2 \to \mathbb{R}^2$$
$$u \to 2u$$
$$v \to v$$

Fix a basis (u, v) of $\mathbb{R}^2$. draw the answer Example 2:

$$T : \mathbb{R}^2 \to \mathbb{R}^2$$
$$u \to u$$
$$v \to 2v$$

draw something Definition: A linear transformation $T : U \to U$ is called scaling if there exists a basis $(u_1, \ldots, u_n)$ of U, $\lambda_1, \ldots, \lambda_n \in k$ such that $T(u_i) = \lambda_i u_i$. How to define T: Proposition: Let $(u_1, \ldots, u_n)$ be a basis of U. $v_1, \ldots, v_n$ arbitrary vectors in V. There exists a unique linear transformation $T : U \to V$ such that $T(u_i) = v_i$. Example:

$$T : U \to V$$
$$u_i \to 0$$

$u_i =$ basis of U. T extends uniquely to U. $T(u) = 0$.

$$[u]_n = \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix}$$

Proof: $u = \lambda_1 u_1 + \cdots + \lambda_n u_n$. $T(u) = \lambda_1 T(u_1) + \cdots + \lambda_n T(u_n) = 0$. Proof: 2 proofs:

1. T exists.
2. T is unique.

Assumptions: $\begin{cases} (u_1, \ldots, u_n) basis \\ T(u_i) = v_i \end{cases}$ Proof of existence: Find T linear such that $T(u_i) = v_i$. Let $u \in U$. Need to define $T(u)$, $u = \lambda_1 u_1 + \ldots + \lambda_n u_n$. Define $T(u) = \lambda_1 T(u_1) + \cdots + \lambda_n T(u_n) \in V$.

1. Prove $T(u_i) = v_i$.
2. T linear.

Proof of $T(u_i) = v_i$: To define T on u_i: coordinates of u in basis $u_i = 0u_1 + \ldots + 1u_i + \ldots + 0u_n$. $T(u_i) = 0v_1 + \cdots + 1v_i + \cdots + 0v_n$.

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Examples: Example 1: The derivative

$$D : P_n \to P_n, \quad f \to f'$$

where $D(x^n) = nx^{n-1}$, Linear map:

- $(f + g)' = f' + g'$
- $(\lambda f)' = \lambda f'$, where $\lambda \in \mathbb{R}$.

Is D injective? No, $D(1) = 0$. $1, 0 \in P_n$, constant polynomials : $r + 0x + 0x^2 + \cdots$. $D(r) = 0$, where $r \in \mathbb{R}$. Is D surjective? No, there is no f such that $f' = 1$. $x^n \notin \Im(D)$. $f \in P_n$ such that $D(f) = x^n$. No sinsce every polynomials n in $\Im(D)$ has degree $\leq n - 1$.

$$D(a_0 + a_1 x + \cdots + a_n x^n) = a_1 + 2a_2 x + \cdots + na_n x^{n-1}.$$

$D : P_n \to P_n$ is not surjective.

Indefinite Integral:

$$\int : P_\infty \to P_\infty$$

[Linear] Injective? $\int 1 = x$, $deg(f) = n$, $deg(\int f) = n + 1$. $\int f = 0$ implies $f = 0$. So injective. $D(\int f) = f$ not surjective. No $f \in P_\infty$ such that $\int f = 1$. Definition: Assume $V = U \oplus U'$, direct sum decomposition of V. $U \cap U' = \{0\}$, $V \in U + U'$. Definition of $P(v)$: Since $V = U \oplus U'$, every $v \in V$ admits a unique sum decomposition $v = u + u'$, by definition $P(v) := u$. $P(v)$ is called the projection of v onto U along U' parallel to U'. Proposition: P is a linear map Proof: $\begin{cases} P(v_1 + v_2) = P(v_1) + P(v_2) \\ P(\lambda v) = \lambda P(v) \end{cases}$

1. Prove : $P(v_1 + v_2) = P(v_1) + P(v_2)$.
2. $v_1 = u_1 + u_1' \in U \oplus U', P(v_1) = u_1$, $v_2 = u_2 + u_2' \in U \oplus U', P(v_2) = u_2$.
3. $P(v_1 + v_2) = ?$, $v_1 + v_2 = u_1 + u_2 + u_1' + u_2'$, $P(v_1 + v_2) = u_1 + u_2 = P(v_1 + v_2)$.

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Kernel and Image: Kernel: $0 \in \ker(T) = \{u \in U : T(u) = 0\}$. Image: $0 \in \Im(T) = \{T(u) : u \in U\} \subseteq V$ Proposition: $\ker(T) \subseteq U$ and $\Im(T) \subseteq V$ are subspaces. Proof: Subspace Criterion Proposition: T surjective $\iff \Im(T) = V$, obvious by definition of surjective. Proposition: T linear $T : U \to V$. Statement: T injective $\iff \ker(T) = \{0\}$, linear system. Kernel is only useful for T linear. $T = \sin$ is non linear, $\ker(sin) = \{x \in \mathbb{R} : sin(x) = 0\}$, not a subspace. Proof: $\ker(T) \neq \{0\}$ then T non-injective $\to$ T injective $\to \ker(T) = \{0\}$. Basically since $[A \implies B] \iff [Not(B) \implies Not(A)]$. Assumption: $\ker(T) \neq \{0\}$ which means $\exists u \in \ker(T), u \neq 0$. Second proof (converse): If T not injective then $\ker(T) \neq \{0\}$. $T(u) = T(u')$. Definition: $\exists u, u' \in U$ s.t $u \neq u'$ and $T(u) = T(u')$. Prove that $T(u) = 0$ for some $v \neq 0$.

$$T(u) - T(u') = 0$$

$$T(u - u') = 0$$

by linearity of T $v \neq 0$. We will show $v := u - u'$ v nonzero and verify $T(v) = 0$. Example: $\Im(P) = U$, $\ker(P) = U'$. $\{v \in V : P(v) = 0\}$. $V = U \oplus U'$, $v = u + u'$, and $P(v) = u \in U$: projection. If $u \in U$ then $P(u) = u$. $\Im(P) = U$: $\Im(P) \subseteq U$: take $U \in \Im(P)$, $u = P(v) \in U$. Then $u = P(v)$ for some V (det of Im). To compute $P(u)$, write $v = U + U'$, $P(v) = u$. $U \subseteq \Im(P)$: $\forall u \in U$, $\exists v : P(v) = u$. $P(v) = u$, take $v = u$, $P(v) = P(u) = u$. $P(v) = u$, $T(u) = 0$.

Example:

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

$$T_A : \mathbb{R}^2 \to \mathbb{R}^2$$

$$T_A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + 2y \\ 3x + 4y \end{pmatrix}$$

Find $\ker(T_A)$.

$$\ker(T_A) = \{\begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : T_A \begin{pmatrix} x \\ y \end{pmatrix} = A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}\}$$

$$\implies \ker(T_A) = \{\begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : \begin{cases} x + 2y = 0 \\ 3x + 4y = 0 \end{cases} \}$$

$\begin{pmatrix} x \\ y \end{pmatrix} \in \ker(T) \iff \begin{pmatrix} x \\ y \end{pmatrix}$ solution to $\begin{cases} x + 2y = 0 \\ 3x + 4y = 0 \end{cases}$. Solve, find $x = y = 0$:

$$\implies \ker(T) = \{\begin{pmatrix} 0 \\ 0 \end{pmatrix}\}$$

which implies T is injective.

Example:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}, \quad T_A : \mathbb{R}^3 \to \mathbb{R}^3$$

$$\ker(T_A) = \{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} : \begin{cases} x + 2y + 3z = 0 \\ 4x + 5y + 6z = 0 \\ 7x + 8y + 9z = 0 \end{cases} \}$$

February 20, 2026 Basis solution: $\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$. And $\ker(T_A) = \text{span}(\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix})$.

Image of a linear transformation $T : U \to V$ is linear, $\Im(T) = \{T(u) : u \in U\}$. Reminder, for a transformation to be linear $T(\lambda u) = \lambda T(u)$, where $\lambda \in k$. Example: $T = T_A$, $T : \mathbb{R}^2 \to \mathbb{R}^2$, $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$. $T(e_1) = Ae_1 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$, $T(e_2) = Ae_2 = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$. $\Im(T) = \text{span}(T(e_1), T(e_2)) = \text{span}(\begin{pmatrix} 1 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \end{pmatrix}) = \mathbb{R}^2$. Proposition: $(e_1, \ldots, e_n)$ basis of U, $\Im(T) = \text{span}(T(e_1), \ldots, T(e_n))$. Main Idea: $\Im(T) = \{T(u) : \forall u \in U\}$, take $u \in U$, meaning $u = \lambda_1 e_1 + \ldots + \lambda_n e_n$, then $T(u) = T(\lambda_1 e_1 + \ldots + \lambda_n e_n) = \lambda_1 T(e_1) + \cdots + \lambda_n T(e_n) \in \text{span}(T(e_i))$. Proof that $\Im(T) \subseteq \text{span}(T(e_1), \ldots, T(e_n))$: Step 1: Let $v \in \Im(T)$, prove $v \in$ span. Step 2: $\exists u$ such that $v \in T(u)$. Show there exist $\lambda_1, \ldots, \lambda_n$ such that $v = \lambda_1 T(e_1) + \cdots + \lambda_n T(e_n)$. Step 3: Find $\lambda_1, \ldots, \lambda_n$, e basis. $\exists \lambda_1, \ldots, \lambda_n$ such that $u = \lambda_1 e_1 + \cdots + \lambda_n e_n$. Apply T: $T(u) = \lambda_1 T(e_1) + \cdots + \lambda_n T(e_n)$.

Matrix of a transformation: dim 1: Proposition: $T : \mathbb{R} \to \mathbb{R}$ linear is determined by a single real (1x1 matrix) number Propostion: $T : \mathbb{R} \to \mathbb{R}$ linear map $T(x) = T(x - 1) = xT(1) - ax$. Matrix of T in basis e: denoted $A = [T]_e$, $[T]_e = ([T(e_1)]_e, \cdots, [T(e_n)]_e) = A$. To compute matrix of $T : U \to U$ in $e =$ basis of U:

1. Compute coord of $T(e_1)$ in basis e.
2. Write these coord as columns in A.

Example: $U = \mathbb{R}^2$, $e =$ std basis,$T : \mathbb{R}^2 \to \mathbb{R}^2$, $[T]_e = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$. Derivative:

February 23, 2026 Matrix of $D(f) = f'$, $D : P_n \to P_n$, $e = (x^n, x^{n-1}, \ldots, x^1, 1)$.

Isomorphisms

Definition 38 (Isomorphism)**.** *A transformation $T : U \to V$ is an isomorphism if*

1. *T is linear.*
2. *T is bijective.*

To check isomorphism:

1. Check T is linear.
2. Check $\ker(T) = \{0\}$.
3. Check $\Im(T) = V$. Basically the vectors $T(e_1), \ldots, T(e_n)$ span V, where $e_1, \ldots, e_n$ is a basis of U.

Theorem 4. *Let $T : V \to V$ is a linear map. A is a matrix in a basis e of V. TFAE:*

1. *T is an isomorphism.*
2. *A is invertible.*
3. *$det(A) \neq 0$.*

Another def of isomorphism: $\exists S$ such that $TS = ST = I$. $T : U \to V$, while $S : V \to U$ and therefore $S = T^{-1}$. For linear $T : U \to V$. If $\dim(U)\ neq \dim(V) \to$ no isomorphism from $U \to V$. We write $U \cong V$ if there exists an isomorphism $T : U \to V$.

Theorem 5. $U \cong V \iff \dim(U) = \dim(V)$. $P_8 \cong \mathbb{R}^9 \cong M_3(\mathbb{R})$.

Corollary 5.1. *Let V be any vector space. If $dim(V) = n$, then $V \cong K^n$.*

Example. *Show x and x^2 are LI in P_2. Solution: $ax + bx^2 = 0 \to a = b = 0$. Proof by isomorphism: transform polynomials into vectors: $\Phi : P_2 \to \mathbb{R}^3$, $f = a + bx + cx^2 \to \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ Just need to prove: $x \to \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, $x^2 \to \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$. Show $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ are linearly independent.*

Example (Could be in exam)**.** *Let two matrices in $M_2(\mathbb{R})$ be $\begin{pmatrix} 2 & 1 \\ 2 & 3 \end{pmatrix}$ and $\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$. Show the matrices are linearly independent. Solve by isomorphism:*

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \to \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}.$$

Basic linear transformation: Let V be a vector space. Let $v_1, \ldots, v_n$ be vectors in V. $\Phi : K^n \to V$ or $\Phi : \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix} \to \lambda_1 v_1 + \cdots + \lambda_n v_n$. $V = Im(\Phi) = span(v_1, \ldots, v_n)$. $Im(\Phi) = \{\lambda_1 v_1 + \cdots + \lambda_n v_n \mid \lambda_1, \ldots, \lambda_n \in K\} = span(v_1, \ldots, v_n)$.

Example: $\Phi : \mathbb{R}^4 \to M_2(\mathbb{R})$, $\Phi : \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 \end{pmatrix} \to \lambda_1 v_1 + \cdots + \lambda_4 v_4$. $v_1 = e_{11}$, $v_2 = e_{12}$, $v_3 = e_{21}$, $v_4 = e_{22}$.

Theorem: $\Phi : K^n \to V$, $\begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix} \to \lambda_1 v_1 + \cdots + \lambda_n v_n = V$. Φ is isomorphic $\iff$ $(v_1, \cdots, v_n)$ is a basis for V. $[v]_e = \Phi^{-1}(v)$. Main basis theorem: $v \in V$, $e = (v_1, \cdots, v_n) \to v_n = \lambda_1 v_1 + \cdots + \lambda_n v_n$. where λ_1 to λ_n are unique coordinates. $[v]_e = \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix}$. Assume $v_1, \cdots, v_n$ basis and Φ injective, $ker(\Phi) = \{0\}$. Assume $\begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix} \in les\Phi$. Prove $\lambda_1, \cdots, \lambda_n = 0$. Know: $\Phi\left(\lambda_1, \vdots, \lambda_n\right) = 0 \to$

February 27, 2026 $\lambda_1 v_1 + \cdots + \lambda_n v_n = 0$.

Review of coordinates: Let $e = (e_1, \cdots, e_n)$ be a basis of V. Any $v \in V$ has coordinates = unique numbers $\lambda_1, \cdots, \lambda_n \in K$, such that $v = \lambda_1 e_1 + \cdots + \lambda_n e_n$. Write $[v]_e = \left(\lambda_1, \vdots, \lambda_n\right)$. Example : $v = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, $[v]_e = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $v = e_1 + e_2$. $f_1 = e_1$, $f_2 = v$ now basis of $\mathbb{R}^2$, $[v]_f = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Example: $f_1 = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$, $f_2 = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$ in $\mathbb{R}^3$ for a basis of a subspace U of $dim2$ in $\mathbb{R}^3$. Define: $[v]_f = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$, $[v]_e =?$, $v = 2f_1 + 3f_2$. Write f_1,f_2 in a matrix: $Q = \begin{pmatrix} 0 & 2 \\ 1 & 1 \\ 2 & 0 \end{pmatrix}$ change of coord matrix. Q takes f coord to e-coord. Definition: Let $e = (e_1, \cdots, e_n)$ be a basis of V. Let $f = (f_1, \cdots, f_k)$ be LI in V. Let $Q = (f_1, \cdots, f_k) \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$ change of basis from f-coord to e-coord. Let V be a vector space, $e = (e_1, \cdots, e_n)$ basis of V. Let $f = (f_1, \cdots, f_k)$ basis of V. Define Q = change of basis f-coord $\to$ e-coord and P = change of basis e-coord $\to$ f-coord. Claim: P, Q are invertible $P = Q^{-1}$. Example: $f_1 = (1, 0)^T$, $f_2 = (1, 1)^T$ compute change of basis matrix, $Q = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, f-coord to e-coor we need $P = Q^{-1}$, need to inverse Q. Q using isomorphism

6 Appendix

6.1 Proof: Standard Basis is a Basis

Claim 1. The vectors $e_1, \ldots, e_n$ span $\mathbb{R}^n$.

Proof. We show that any vector in $\mathbb{R}^n$ can be written as a linear combination of $e_1, \ldots, e_n$:

$$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} x_1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ x_2 \\ \vdots \\ 0 \end{pmatrix} + \cdots + \begin{pmatrix} 0 \\ \vdots \\ 0 \\ x_n \end{pmatrix} = x_1 e_1 + x_2 e_2 + \cdots + x_n e_n.$$

□

Claim 2. The vectors $e_1, \ldots, e_n$ are linearly independent.

Proof. Suppose

$$\lambda_1 e_1 + \cdots + \lambda_n e_n = 0.$$

Then

$$\begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix},$$

so $\lambda_1 = \cdots = \lambda_n = 0$. Therefore $e_1, \ldots, e_n$ are linearly independent. □

6.2 Algebraic Structures: Rings, Fields, and Vector Spaces

Rings vs Vector Spaces. Although rings and vector spaces both have addition and a zero element, they differ fundamentally in how multiplication works:

- **Ring:** A set R with addition and multiplication between elements of R. Addition forms an abelian group, multiplication is associative, and distributes over addition. Scalars are elements *inside the set*.
- **Vector Space:** A set V with addition and multiplication by scalars from a *field* F. Addition forms an abelian group, scalar multiplication distributes appropriately. There is no multiplication between vectors themselves; only scalar multiplication is defined.

7 Solutions

Solution 1 (Hermitian Matrix)**.** *This follows by the rule* $(AB)^T = B^T A^T$ *and* $\overline{AB} = \overline{A}\,\overline{B}$*:*

$$(AB)^* = (\overline{AB})^T = (\overline{A}\,\overline{B})^T = \overline{B}^T \overline{A}^T = B^* A^*.$$

Solution 2 (Invertibility)**.** *Suppose B and B' are both inverses of A. Then*

$$B = BI = B(AB') = (BA)B' = IB' = B'.$$

Therefore, $B = B'$, so the inverse is unique.

Solution 3 (Invertibility 2)**.** *We can answer this problem with proof by contradiction. Let's suppose this matrix is invertible. By definition there exists $B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ such that $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. We can rewrite this equation into: $\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}^{-1}$. The inverse of our matrix can be rewritten as $\frac{1}{0*0-1*0} \begin{bmatrix} 0 & -1 \\ 0 & 0 \end{bmatrix}$[12]. But this is undefined since division by 0 is undefined. Therefore, our initial assumption that the matrix is invertible is false, and thus the matrix is not invertible.*

Solution 4 (Span)**.**

$$\operatorname{span}(v_1, v_2) = \{xv_1 + yv_2 : x, y \in \mathbb{R}\} = \left\{ \begin{pmatrix} 3x + y \\ x + 3y \end{pmatrix} : x, y \in \mathbb{R} \right\}.$$

Solution 5 (Span 2)**.** *Assume $u = au_1 + bu_2 + cu_3$, with $a, b, c \in \mathbb{R}$. This gives the system of equations*

$$\left(\begin{array}{ccc|c} 3 & 1 & 2 & 2 \\ 10 & 3 & 8 & 10 \\ 7 & -2 & 1 & 7 \end{array}\right).$$

Solving via Gaussian elimination, we find $a = \frac{2}{21}$, $b = -\frac{46}{21}$, and $c = \frac{41}{21}$. Hence $u \in \operatorname{span}(u_1, u_2, u_3)$, meaning that u is in the span of u_1, u_2, u_3.

Solution 6 (Subspace Criterion)**.** *Let $A \in M_n(k)$ be fixed and define*

$$U = \{x \in k^n : Ax = \vec{0}\}.$$

We verify the subspace criterion.

(0) Non-empty: Since $A0 = 0$, we have $0 \in U$.

(1) Closed under addition: Let $x, y \in U$. Then $Ax = 0$ and $Ay = 0$. Hence

$$A(x + y) = Ax + Ay = 0 + 0 = 0,$$

so $x + y \in U$.

(2) Closed under scalar multiplication: Let $x \in U$ and $\lambda \in k$. Then

$$A(\lambda x) = \lambda Ax = \lambda 0 = 0,$$

so $\lambda x \in U$.

Therefore, by the subspace criterion, U is a subspace of K^n. It is called the null space (kernel) of A.

[12] Recall that an inverse of a 2×2 matrix is equal to its determinant multiplied with its conjugate

Solution 7 (Subspace Criterion 2)**.** *We verify the subspace criterion.*

(0) Non-empty: Since $A0 = 0$*, we have* $0 \in U$*.*

(1) Closed under addition: Let $y_1, y_2 \in U$*. Then there exist* $x_1, x_2 \in k^n$ *such that*

$$y_1 = Ax_1, \quad y_2 = Ax_2.$$

Hence,

$$y_1 + y_2 = Ax_1 + Ax_2 = A(x_1 + x_2) \in U.$$

(2) Closed under scalar multiplication: Let $y \in U$ *and* $\lambda \in k$*. Then* $y = Ax$ *for some* $x \in k^n$*, and*

$$\lambda y = \lambda Ax = A(\lambda x) \in U.$$

Therefore, by the subspace criterion, U *is a subspace of* k^n*. It is called the image of* A*.*

Solution 8 (Subspace Criterion 3)**.** *Let* $V \subseteq k$ *be a subspace. We show that either* $V = \{0\}$ *or* $V = k$*. If* $V = \{0\}$*, we are done. Otherwise,* $V \neq \{0\}$*. Then there exists* $v \in V$ *with* $v \neq 0$*. We prove that* $V = k$*. Let* $x \in k$ *be arbitrary. Since* $v \neq 0$*, there exists* $\lambda \in k$ *such that*

$$x = \lambda v.$$

Because V *is closed under scalar multiplication,* $\lambda v \in V$*, hence* $x \in V$*. Therefore every* $x \in k$ *belongs to* V*, so* $V = k$*. Conclusion: the only subspaces of* k *are* $\{0\}$ *and* k*.*

Solution 9 (Subspace Criterion 4)**.** *We verify the subspace criterion. (0) Non-empty: Taking* $\lambda_1 = \cdots = \lambda_n = 0$ *gives*

$$0 = 0v_1 + \cdots + 0v_n \in \operatorname{span}(v_1, \ldots, v_n).$$

(1) Closed under addition: Let $u, v \in \operatorname{span}(v_1, \ldots, v_n)$*. Then there exist scalars* $a_1, \ldots, a_n, b_1, \ldots, b_n \in k$ *such that*

$$u = a_1v_1 + \cdots + a_nv_n, \quad v = b_1v_1 + \cdots + b_nv_n.$$

Hence,

$$u + v = (a_1 + b_1)v_1 + \cdots + (a_n + b_n)v_n \in \operatorname{span}(v_1, \ldots, v_n).$$

(2) Closed under scalar multiplication: Let $u \in \operatorname{span}(v_1, \ldots, v_n)$ *and* $\lambda \in k$*. Then*

$$u = a_1v_1 + \cdots + a_nv_n$$

for some scalars a_i*, and*

$$\lambda u = (\lambda a_1)v_1 + \cdots + (\lambda a_n)v_n \in \operatorname{span}(v_1, \ldots, v_n).$$

Therefore, $\operatorname{span}(v_1, \ldots, v_n)$ *is a subspace of* V*.*

Solution 10 (Span Membership)**.** *We look for scalars $x, y, z \in \mathbb{R}$ such that*

$$xv_1 + yv_2 + zv_3 = v.$$

That is,

$$x\begin{pmatrix}1\\2\\1\end{pmatrix} + y\begin{pmatrix}2\\5\\4\end{pmatrix} + z\begin{pmatrix}1\\3\\6\end{pmatrix} = \begin{pmatrix}3\\5\\-5\end{pmatrix}.$$

This gives the system

$$\begin{cases} x + 2y + z = 3, \\ 2x + 5y + 3z = 5, \\ x + 4y + 6z = -5. \end{cases}$$

Solving, we obtain

$$x = 3, \quad y = 1, \quad z = -2.$$

Therefore,

$$v = 3v_1 + v_2 - 2v_3,$$

so $v \in \operatorname{span}(v_1, v_2, v_3)$.

Solution 11 (Span Membership 2)**.** *We seek scalars $t, y, z \in \mathbb{R}$ such that*

$$f = tf_1 + yf_2 + zf_3.$$

Comparing coefficients,

$$t(x^2 + 2x + 1) + y(2x^2 + 5x + 4) + z(x^2 + 3x + 6) = 3x^2 + 5x - 5,$$

which gives

$$\begin{cases} t + 2y + z = 3, \\ 2t + 5y + 3z = 5, \\ t + 4y + 6z = -5. \end{cases}$$

Solving,

$$t = 3, \quad y = 1, \quad z = -2.$$

Hence,

$$f = 3f_1 + f_2 - 2f_3,$$

and therefore $f \in \operatorname{span}(f_1, f_2, f_3)$.

Solution 12 (Linearly independent vectors)**.** *x, y, z not all zero.*

$$x\begin{pmatrix}1\\1\\0\end{pmatrix} + y\begin{pmatrix}1\\3\\2\end{pmatrix} + z\begin{pmatrix}4\\9\\5\end{pmatrix} = \begin{pmatrix}0\\0\\0\end{pmatrix}$$

$$\begin{cases} x + 3y + 4z = 0 \\ 1 + 3y + 9z = 0 \\ 0 + 2y + 5z = 0 \end{cases}$$

Find a non-trivial solution. $x = 3$, $y = 5$, $z = -2$. Not LI: $3v_1 + 5v_2 - 2v_3 = 0$.

8 Useful Links

Proofs: https://www.youtube.com/watch?v=V5tUc-J124s